Class 12

Parisa Charoensook A17939591

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Identify Genetic Variants of Interest

> Q1: What are those 4 candidate SNPs?

The four candidate SNPs are rs12936231, rs8067378, rs9303277, and rs7216389

> Q2. What three genes do these variants overlap or effet?

The three genes that these variants overlap and affect aew PBP2, GSDMB, ORMDL3.

> Q3: What is the location of rs8067378 and what are the different alleles for rs8067378?

rs8067378 is located on chromosome 17:39895095, and has the alleles A and G.

> Q4: Name at least 3 downstream genes for rs8067378?

Three downstream genes are GSDMB, ORMDL3, and IKZF3.

> Q5. What proportion of the Mexican Ancestry in Los Angeles sample population (MXL) are homozygous for the asthma associated SNP (G|G)?

```
geno <- read.csv ("table.csv")

head(geno)
```

```
  Sample..Male.Female.Unknown. Genotype..forward.strand. Population.s. Father
1                  NA19648 (F)                       A|A ALL, AMR, MXL      -
2                  NA19649 (M)                       G|G ALL, AMR, MXL      -
3                  NA19651 (F)                       A|A ALL, AMR, MXL      -
4                  NA19652 (M)                       G|G ALL, AMR, MXL      -
5                  NA19654 (F)                       G|G ALL, AMR, MXL      -
6                  NA19655 (M)                       A|G ALL, AMR, MXL      -
  Mother
1      -
2      -
3      -
4      -
5      -
6      -
```

```r
table(geno$Genotype)
```

```
A|A A|G G|A G|G
 22  21  12   9
```

```r
counts <- table(geno$Genotype)
counts["G|G"] / sum(counts)
```

```
     G|G
0.140625
```

The proportion of the MXL population that is homozygous for asthma-associated SNP is 9/64 or ~14.1%.

> Q6. Back on the ENSEMBLE page, use the "search for a sample" field above to find the particular sample HG00109. This is a male from the GBR population group. What is the genotype for this sample?

The genotype sample HG00109 is G|G.

Section 2: Intial RNA - Seq analysis

> Q7: How many sequences are there in the first file? What is the file size and format of the data? Make sure the format is fastqsanger here!

The first file has 3863 sequences. The file size is 757 KB and its format is fastqsanger.

> Q8: What is the GC content and sequence length of the second fastq file?

The GC content is 54% and the sequence length is 50-75 bp on the second fastq file.

> Q9: How about per base sequence quality? Does any base have a mean quality score below 20?

The per base sequence qualtiy score remains above 20 throughout the reads, no bases have a mean quality score below 20. ## Section 3: Mapping RNA - Seq reads to genome.

> Q10: Where are most the accepted hits located?

The most accepted hits are located on chromosome 17 around chr17:38007296–38170000.

> Q11: Following Q10, is there any interesting gene around that area?

Genes like EZH1, CNTNAP1, and CCR10.

> Q12: Cufflinks again produces multiple output files that you can inspect from your right-handside galaxy history. From the "gene expression" output, what is the FPKM for the ORMDL3 gene? What are the other genes with above zero FPKM values?

The FPKM value for the ORMDL3 gene was 0. Based on gene expression output, no other genes showed above zero FPKM values either.